



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru

BTech VI Semester

End Semester Examination - April / May -2025

SYSTEM ON CHIP [ECE-3222]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) You are required to develop a Set-Top Box SoC which inputs a signal and transforms the data to content displayed on a TV screen. Consider a wafer yield of 98%, and a defect density of 0.1 per square cm. The target feature size is at 65nm technology node. The Set-Top Box SoC architecture must consist of the following components:

Unit	Area (A)
64b CISC processor with 48KB of I-cache and 32KB of D-cache	300
A Texas Instrument DSP (for video signal acquisition)	150
A SHARC-based DSP (for demodulation/ error correction schemes)	185
MPEG-2 transport stream de-multiplexer accelerator unit	215
Bus and bus control	400
Application Memory (512KB)	1024
Shared SRAM L2 cache	

(5)

Fig. 1a

Assume 12% chip area for net die costs i.e. IO pads etc, and 13% overhead integration for connecting the shared L2 cache to the rest of the components in the SoC. Generate an architecture and floorplan for the SoC. Show all calculations and the steps undergone to implement your design. Specify all component areas, and the final capacity of SRAM L2 cache your system may support. (For feature size of 65nm technology, number of A per mm² is 236.69)

- B) Design a finite state machine for a serial binary input which is divisible by 4. (3)
- C) What is pipelining in a processor? What are different hazards which can be encountered during pipelining? (2)
- 2) Write a pseudo code for implementing factorial of a given number and draw a Datapath for the same. (5)
- A)
- B) Explain the SoC development life cycle with a neat figure. (3)
- C) Explain the significance of Amdahl's Law with necessary equations. (2)
- 3) A certain pipeline has the following functions and functional unit delays (without clocking overhead) as shown in Fig 3a: (4)

A)

Function	Delay
A	0.6
B	0.8
C	0.3
D	0.7
E	0.9
F	0.5

Fig.3a

Function units B, D and E can be subdivided into two equal delay stages. If the expected occurrence of pipeline breaks is $b=0.3$ and clocking overhead is $0.1ns$:

- What is the optimum number of pipeline segments (round down to integer value)?
- For the calculated segments in (a), what is the cycle time that maximizes performance without allocating multiple cycles to a segment?
- Compute the pipeline performance with this cycle time.

- B) Use K-mean algorithm to cluster the following 7 examples into 2 clusters: $A1=(189,72)$, $A2=(170,56)$, $A3=(178,60)$, $A4=(149,68)$, $A5=(182,72)$, $A6=(188,77)$, $A7=(180,71)$. (4)

- C) Explain the need for NoC with respect to bus based communication. (2)

- 4) What is arbitration? Explain the difference between priority based and TDMA based arbitration protocols. (5)

A)

- B) What is the significance of routing in NoC? Explain anyone routing algorithms of NoC. (3)

- C) Differentiate between 2-D mesh and torus topologies of NoC. (2)

- 5) Find the minimum number of Test vectors to detect stuck at faults at the given points : a, b, c, d, f of the circuit shown in Fig.5a

A)

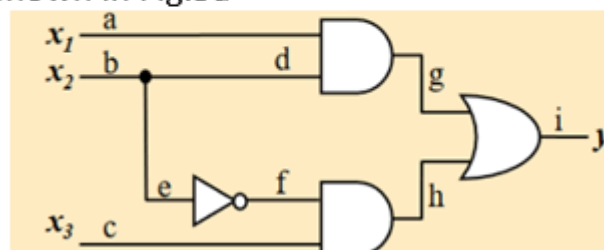


Fig.5a

- B) Explain the fish bone clock structure and Pi tree clock structure with neat diagram. (3)

- C) Classify the IP's based on size and explain them. (2)

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